Homework 9

> NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Read Chapters 2-3, Beej's Guide to Network Programming.

2. Explain the primary differences between `SOCK_STREAM` and `SOCK_DGRAM`. In what scenario would an application developer choose a “connectionless” socket over a “reliable (connection-oriented)” one?

3. A socket is represented by a file descriptor. Explain how the Linux kernel treats a network socket similarly to a standard file on disk, and identify one system call that works on both.

4. Define the difference between Big-Endian and Little-Endian byte orders. If a host machine uses Little-Endian internally, why must a programmer use `htons()` before sending a port number over the network? What would happen if this step were skipped?

5. Contrast IPv4 and IPv6 addresses in terms of bit length and human-readable notation.

6. Explain the concept of a subnet mask. How does a computer use a mask like 255.255.255.0 to determine if a destination IP is on the local network or requires a gateway?

7. Define the role of an ephemeral port and identify which side of the connection (client or server) typically uses it. Contrast this with well-known ports and provide two examples of specific services and their associated port numbers.

8. Given a client at 128.2.194.242 using port 51213 and a server at 208.216.181.15 using port 80, write out the unique socket pair that identifies their connection.

- **Welcome: Sean**
  - LogOut

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